An Analysis of Geographic Patterns in Building Inspection Deficiency and Inspection Failures in Calgary.

Masters of Data Analytics (MDA)

Group 1

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Abstract

The City of Calgary inspects every new residential building at several stages of construction, recording each visit as a pass or a non-pass against the Alberta Building Code. This project asks a simple question with practical consequences: are inspection failures concentrated in specific quadrants of the city? Using the relational dataset provided with 107,067 inspections spanning 2000 to 2025, we extracted a clean analytical dataset with SQL, explored it, visualized it, and then tested the geographic hypothesis formally using a Chi-Square test and three classification models; logistic regression, decision tree, and a random forest.

The result shows that geography barely matters and failure rates differ slightly across the four zones with NW having the highest rate at 38.1% and NE the lowest at 36.3%. Once we controlled for inspection type and year, no zone showed meaningful or higher odds of failure, and zone contributes almost nothing to predictive accuracy. What mattered the most was the inspection type; gas and fire protection inspections fail more than 80% of the time, against under 10% for foundations. Our position remains that the city allocates inspection by the type of job rather than where it happens.

Chapter 1: Introduction

1.1 Overview of the Data

The Calgary Building Inspections data provided is a relational database with records of residential building permits issued, inspections and code violations (called deficiencies) in the City of Calgary, Alberta from 2000 to 2025. In Calgary before construction commences, every new residential building is required to obtain a permit while city inspectors visit the site at different stages of the project to assess jobs done by contractors against the city code of standards. During these inspections, any violations found are recorded as deficient. These deficiencies are required to be fixed by the contractor or permit holder before construction can proceed or for the permit to be closed.

The geographic data provided forms the foundation of this project because it makes it possible to examine deficiency rates and inspection outcomes across the city and determine whether certain areas show higher rates of violations or failures.

1.2 Description of the Database Structure

The database is organized using a relational structure in which fact tables are connected to dimensional tables that convert numerical categories into readable labels.

Table	What It Contains	Key fields
Inspection	One row refers to every inspection visit. Stores the permit it belongs to, the inspection type, the outcome, and the date it was completed.	inspection_id, permit_id, inspection_type_id, outcome_id, inspection_completed_date
Dim_outcome	Lookup table that shows outcomes as Acceptable or Not Acceptable.	outcome_id, outcome_desc
Dim_inspection_type	Lookup table that classifies inspection into Foundation, Framing or Gas.	inspection_type_id, inspection_type_desc
Permit	One row per building permit. Functions as a bridge to connect each inspection to its physical location.	permit_id, location_id
Location	Stores the address and GPS coordinates of each permit site.	location_id, address_line, latitude, longitude

The join path that connects all five tables together has been modeled in chapter 4 to help have a better understanding of how each table is related.

How outcomes are classified

The dim_outcome table contains 20 different outcomes. For this project, we grouped all outcomes into three categories Pass, Non-pass and Excluded. If an inspection outcome is acceptable, such inspection will pass with no or minor issues. If the outcome is unacceptable, such an inspection will not pass. If the outcome is not required, inspection is excluded or will not take place.

1.3 Data Exploration using SQL and Preliminary Observation

Query 1: How many records are in the main tables?

Purpose: Summary count of relevant tables in our data.

Our findings from the query revealed a total of 78,394 permits, 107,067 inspections were carried out in 71,771 locations with 123,802 inspection deficiency recorded. This count validates our population before any further analysis begins.

Data Exploration SQL Query

```
SELECT “permit” AS data_info, COUNT(*) AS total FROM permit
UNION ALL
SELECT 'inspection' AS data_info, COUNT(*) AS total FROM inspection
UNION ALL
SELECT 'location' AS data_info, COUNT(*) AS total FROM location
UNION ALL
SELECT 'inspection_deficiency' AS data_info,
       COUNT(*) AS total FROM inspection_deficiency;
```

Query 2: How many inspections are in each zone?

Purpose: We categorized inspections by the permit address suffix to show how the data is distributed geographically. We use the CASE statement to read the last characters of every address_line field to assign each inspection to a zone.

Our findings show that SE has a total of 34029 inspections, SW has 29268 inspections, NW has 26136 inspections, and NE has 17634 inspections reflecting where most residential constructions are taking place. This serves as the project's first geographic comparison of inspection activity across Calgary. The higher inspection volume observed in the SE and SW zones may reflect increased residential development activity in those areas during the study period.

SQL Query Used to Count Inspections Across Zones (NW, NE, SW, SE)

```
SELECT
```

```
    CASE
        WHEN l.address_line LIKE '% NW' THEN 'NW'
        WHEN l.address_line LIKE '% NE' THEN 'NE'
        WHEN l.address_line LIKE '% SW' THEN 'SW'
        WHEN l.address_line LIKE '% SE' THEN 'SE'
        ELSE 'Other'
    END AS zones,
    COUNT(*) AS total_inspections
FROM inspection i
JOIN permit p ON i.permit_id = p.permit_id
JOIN location l ON p.location_id = l.location_id
GROUP BY zones
ORDER BY total_inspections DESC;
```

Query 3: What is the failure rate per zone?

Purpose: For our exploratory analysis, this query would calculate the proportion of inspections that resulted in non-pass for each zone. The result will be an early indication if truly geographic differences in failure rates exist before any formal statistical testing is done.

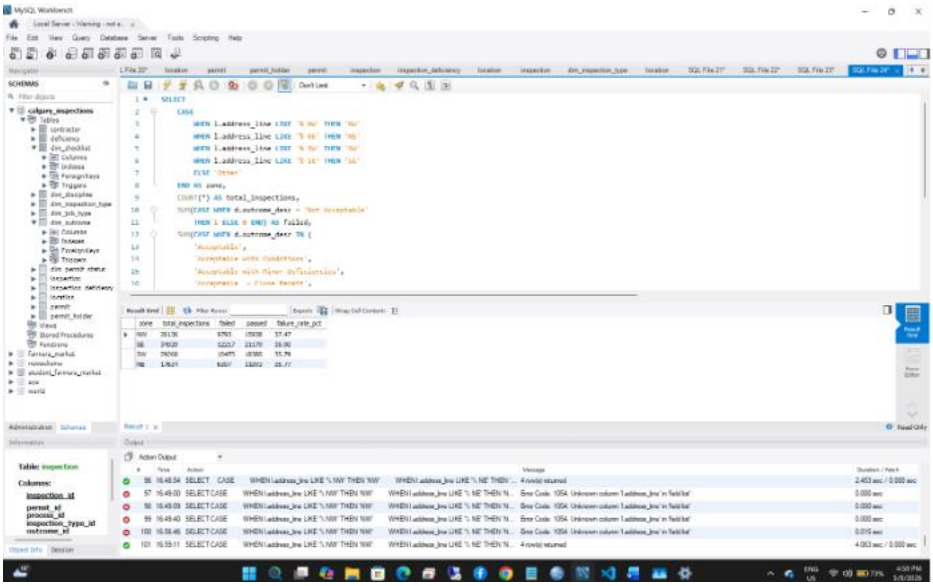
zone	total_inspections		failed	failure_rate_pct %
NW	26136		9793	37.47
SE	34029		12217	35.9
SW	29268		10475	35.79
NE	17634		6307	35.77

Outcome: Our findings revealed that failure rates differ slightly across zones. However, the differences between zones are relatively small, suggesting that geographic location alone may not fully explain inspection outcomes. Other factors such as inspection type, project complexity, or contractor practices may also influence failure rates. Further statistical testing will be conducted later in this study.

SQL Query (Failure Rate Per Zone)

```
SELECT
    CASE
        WHEN l.address_line LIKE '% NW' THEN 'NW'
        WHEN l.address_line LIKE '% NE' THEN 'NE'
        WHEN l.address_line LIKE '% SW' THEN 'SW'
        WHEN l.address_line LIKE '% SE' THEN 'SE'
        ELSE 'Other'
    END AS zone,
    COUNT(*) AS total_inspections,
    SUM(CASE WHEN d.outcome_desc = 'Not Acceptable'
        THEN 1 ELSE 0 END) AS failed,
    SUM(CASE WHEN d.outcome_desc IN (
        'Acceptable',
```

```
        'Acceptable with Conditions',
        'Acceptable with Minor Deficiencies',
        'Acceptable  - Close Permit',
        'Acceptable Add Inspection')
        THEN 1 ELSE 0 END) AS passed,
    ROUND(
        SUM(CASE WHEN d.outcome_desc = 'Not Acceptable'
            THEN 1 ELSE 0 END)
        / COUNT(*) * 100, 2) AS failure_rate_pct
FROM inspection i
JOIN permit p ON i.permit_id = p.permit_id
JOIN location l ON p.location_id = l.location_id
JOIN dim_outcome d ON i.outcome_id = d.outcome_id
GROUP BY zone
ORDER BY failure_rate_pct DESC;
```



Query 4: Do we have missing GPS coordinates?

Purpose: We had SQL run a query to check for missing latitude and longitude values for all location records as this would be important for our heatmap visualization.

Outcome: Both missing_lat and missing_long returned zero indicating that GPS data is complete, and all permit locations can be visualized.

SQL Query (Missing GPS Coordinates)

```
SELECT
    COUNT(*) AS total_locations,
    SUM(CASE WHEN latitude IS NULL
        THEN 1 ELSE 0 END) AS missing_lat,
    SUM(CASE WHEN longitude IS NULL
```

```
        THEN 1 ELSE 0 END) AS missing_long
FROM location;
```

Query 5: Failure Rate by Inspection Type

Purpose: This helps us ascertain if different inspection types have different failure rates. This query will provide evidence as to why inspection type must be controlled for the main analysis.

Outcome: Our findings revealed that failure rates differ significantly across inspection types. But are these differences significant statistically? We shall be investigating this using the appropriate statistical tool. The very high failure rates observed in Gas Final and Gas Rough inspections may reflect stricter safety requirements and the technical complexity associated with gas-related construction work. In contrast, Foundation inspections showed much lower failure rates, suggesting fewer compliance issues during early-stage construction.

Result

inspection_type_desc	total_inspections	failed	failure_rate_pct
Gas Final	1851	1516	81.9
Gas Rough	3899	3178	81.51
FP Rough	1436	1080	75.21
Final	21535	9786	45.44
FP Final	19	8	42.11
Temp Gas	54	21	38.89
Framing	49785	18862	37.89
Occupancy	3881	1333	34.35
Pub Protect Safety	3	1	33.33
Rough	2428	625	25.74
Underground	635	147	23.15
Progress	9712	1188	12.23
Foundation	11804	1045	8.85
Site Safety	25	2	8

SQL Query (Failure Rate by Inspection)

```
SELECT
    t.inspection_type_desc,
    COUNT(*) AS total_inspections,
    SUM(CASE WHEN d.outcome_desc = 'Not Acceptable'
        THEN 1 ELSE 0 END) AS failed,
    ROUND(
        SUM(CASE WHEN d.outcome_desc = 'Not Acceptable'
            THEN 1 ELSE 0 END)
        / COUNT(*) * 100, 2) AS failure_rate_pct
FROM inspection i
```

```
JOIN dim_inspection_type t
    ON i.inspection_type_id = t.inspection_type_id
JOIN dim_outcome d ON i.outcome_id = d.outcome_id
GROUP BY t.inspection_type_desc
ORDER BY failure_rate_pct DESC;
```

1.4 Data Quality Issues, Assumptions and Limitations.

Data Quality issues

- **Some outcome values do not represent genuine inspections:** The dim_outcome table contains values such as Not Required, Incorrectly Called For, and CSR Skipped. These are administrative values and they do not represent actual inspection outcomes. As such, have been excluded from all pass and non pass calculations.
- **The permit_deficiencies_flag column is always zero:** This field in the inspection table shows 0 for every record in the dataset. It cannot be used to identify failed inspections or the presence of deficiencies. All outcome information must come from the outcome_id joined to dim_outcome.
- **Contractor data is mostly missing:** The permit table has a contractor_id column but most records do not have a value filled in. Contractor-level comparisons are therefore not possible and are excluded from the scope of this project.

Assumptions

- Outcomes classified as Acceptable, Acceptable with Conditions, Acceptable with Minor Deficiencies, and Acceptable Close Permit are treated as pass outcomes.
- Outcomes classified as Not Acceptable, No Entry, Unable to Enter, and Stop Work are treated as non-pass outcomes.
- Outcomes such as Not Required, Incorrectly Called For, and Cancelled are excluded because they do not represent a genuine inspection result.
- Address zones are derived from the last two characters of the address_line field in the location table and is used as the geographic grouping variable.
- GPS coordinates in the location table are assumed to be accurate and complete, as confirmed through Query 4.

Limitations

- No income or land use data is available in this database, so the analysis cannot explain why geographic differences exist only if they do.

- The main geographic comparison uses an address suffix, which is a broad grouping. Two permits in the same zones can be in different neighborhoods with different characteristics.
- Data provided was from 2000 to 2025, but inspections are concentrated between 2005 – 2023 which simply means findings might not be equally represented across all years.

1.5 Preliminary Observations

After running the exploratory queries above, three observations stood out that were of interest and subsequently served as a direction for our research established in Chapters 2 and 3.

- **Inspections are present in all four Zones:** Query 2 confirms that records exist in NW, NE, SW and SE Calgary, meaning all four groups can be measured and compared against each other.
- **GPS coordinates are complete and usable:** Query 4 confirms that latitude and longitude values are present for all location records with no missing values. This means the heatmap visualization in other parts of this project will be produced without any gaps in the geographic data.
- **Failure rates vary by inspection type:** Query 5 shows that failure rates differ across inspection types. This validates our reason for including inspection type as a control variable as zones with more gas or framing inspections could appear to have a higher failure rate simply because of the type of work being inspected, not because of its location.

Chapter 2: Problem Statement and Research Question

2.1 Problem Statement

The City of Calgary conducts building inspections to ensure that new construction and renovation work meets the standards set out in the Alberta Building Code. When a site visit reveals work that does not meet these standards, the inspection is recorded as a failure, and the contractor is required to correct the problem before construction can continue.

Inspection failures happen across the city, but it is not currently known whether certain areas experience consistently higher failure rates than others. If geographic patterns do exist, they could point to differences in construction complexity, contractor practices, or the volume of development happening in a particular area at a given time. Understanding where failures are concentrated is important because it can help the city allocate inspection resources better, target training where it is needed most, and improve building code compliance across all parts of Calgary.

Our aim is to quantify if inspection failure rates differ significantly across the four geographic zones of Calgary (NW, NE, SW, and SE) and identify which areas, if any, are associated with higher rates of failed inspections.

2.2 Research Question

Do building inspection deficiencies and failed inspections in Calgary show geographic patterns, and are certain areas of the city associated with higher inspection failure rates than others?

2.3 Research Scope

This research investigates inspection outcomes across all four geographic zones of Calgary using the address suffix available in every permit record. The proportion of inspections that result in a non-pass outcome is calculated for each zone and compared to determine whether geographic location is associated with a higher or lower likelihood of inspection failure. GPS coordinates from the location table are used as a supplementary tool to visualize the geographic distribution of failures through a heatmap.

Key elements included

- All inspections with a recorded outcome_id in the dataset.
- Geographic grouping based on the directional address suffix (NW, NE, SW, SE) derived from the address_line field in the location table.
- GPS Data provided (latitude and longitude) from the location table would be used for heatmap visualization.
- Compare inspection failure rates across the four zones.
- Inspection type as a control variable to account for the fact that different types of inspections may have naturally different pass and fail rates.

2.4 Importance of Study

This research is relevant to individuals, groups, and government (institutions and policy makers) that have an interest in Calgary's residential construction sector.

- **For the City of Calgary:** our findings would shed more light into areas with high inspection failure rates. This would help the city decide on where to focus on enforcement activity, how to allocate human resources, and where training might be most beneficial.
- **For investors and homebuyers:** Information on areas with lower inspection failure rates helps reduce investment risk and provides a clearer picture of construction quality across different parts of the city.
- **For the research community:** This research demonstrates how building inspection data, combined with geographic grouping and GPS coordinates, can be used to identify patterns in construction compliance.

Chapter 3: Research Hypothesis

3.1 Research Hypothesis

In defining our Research question based on the data provided and preliminary assessment done, we were able to formulate a clear hypothesis that inspection failure rates may vary across the different geographic areas identified in the dataset and areas with higher construction activity could experience more failures.

H_0 (Null Hypothesis)

There is no significant difference in inspection failure rate across the four geographic zones of Calgary (NW, NE, SW, and SE).

H_1 (Alternate Hypothesis)

There is a significant difference in inspection failure rates across the four geographic zones of Calgary (NW, NE, SW, and SE).

3.2 Variables and Proposed Test

Dependent variable: Inspection failure rate (Pass vs. Non-Pass): This will be measured as both the count and proportion of failed inspections per zone. An inspection is classified as a non-pass if its outcome_desc is Not Acceptable, No Entry, Unable to Enter, or Stop Work. Outcomes with no genuine inspection results such as Not Required or Cancelled are excluded.

Independent variable: Geographic zones (NW, NE, SW, SE): A categorical grouping variable was derived from the directional address suffix in the location table. Each inspection is assigned to a zone based on the last two characters of the address_line field.

Control variable Inspection type: Several inspection types such as foundation, framing, gas or final inspections may have different pass and fail rates regardless of location. If certain zones have a higher concentration of technical inspection types, this could influence its failure rate independently from geographical factors.

Statistical test: To ascertain if inspection failure rates differ significantly across all zones, a Chi-Square Test of Independence will be used as the primary test. This is appropriate because both the outcome variable (pass or non-pass) and the grouping variable (zones) are categorical. A significance level of $p < 0.05$ will be used.

To account for the control variable, a logistic regression model will be used with geographic zone as the main predictor and inspection type included as a control variable to estimate the odds of failed inspection for each zone while holding inspection types constant.

3.3 Significance Level and Decision Rule

If the null hypothesis is not rejected ($p \geq 0.05$): Then the data suggests that building inspection failure rates are roughly the same across NW, NE, SW and SE and any observed differences are within the range of chance variation.

If the null hypothesis is rejected ($p < 0.05$): Then the data indicates that at least one zone has a statistical difference in failure rate compared to the others.

It is worth noting that both results are valid as our findings aim to solve our research questions through answers provided by our data.

The statistical findings suggest that while geography may influence inspection outcomes, inspection type and construction complexity are also important contributing factors.

Chapter 4 Data Collection and SQL Queries

This chapter documents SQL queries used to extract data needed to address our research question. We ran all queries on MySQL Workbench against the Calgary Building Inspections database and exported our results to a CSV file. We used Python Libraries for all subsequent analysis, modelling, and visualization.

4.1 Data Collection Method

For our data collection, we extracted Address Suffix as Zone, we used the SQL binary encoding to categorize inspection outcomes (Pass =0, Non-Pass=1). We also exported data of inspection type carried out and GPS Coordinates. Using the SQL JOIN function below, we were able to query data from different tables within the database to produce our dataset for analysis. Inspections with administrative outcomes such as Not Required, incorrectly called for, and cancelled were excluded from our data extraction.

SQL Data Extraction Query

```
SELECT
    i.inspection_id,
    i.inspection_completed_date,
    YEAR(i.inspection_completed_date)   AS inspection_year,

    -- Geographic zone derived from address suffix
    CASE
        WHEN l.address_line LIKE '% NW' THEN 'NW'
        WHEN l.address_line LIKE '% NE' THEN 'NE'
        WHEN l.address_line LIKE '% SW' THEN 'SW'
        WHEN l.address_line LIKE '% SE' THEN 'SE'
        ELSE 'Other'
    END                                 AS zone,

    -- GPS coordinates for heatmap visualization
    l.latitude,
    l.longitude,
    l.address_line,

    -- Inspection type (control variable)
    t.inspection_type_desc,

    -- Raw outcome label
    d.outcome_desc,

    -- Binary outcome: 1 = Non-Pass, 0 = Pass
    CASE
        WHEN d.outcome_desc IN (
            'Not Acceptable',
            'No Entry',
```

```
            'Unable to Enter',
            'Stop Work')
        THEN 1
        WHEN d.outcome_desc IN (
            'Acceptable',
            'Acceptable with Conditions',
            'Acceptable with Minor Deficiencies',
            'Acceptable - Close Permit',
            'Acceptable Add Inspection')
        THEN 0
        ELSE NULL
    END                                 AS failed

FROM inspection i
JOIN permit p
    ON i.permit_id = p.permit_id
JOIN location l
    ON p.location_id = l.location_id
JOIN dim_outcome d
    ON i.outcome_id = d.outcome_id
JOIN dim_inspection_type t
    ON i.inspection_type_id = t.inspection_type_id

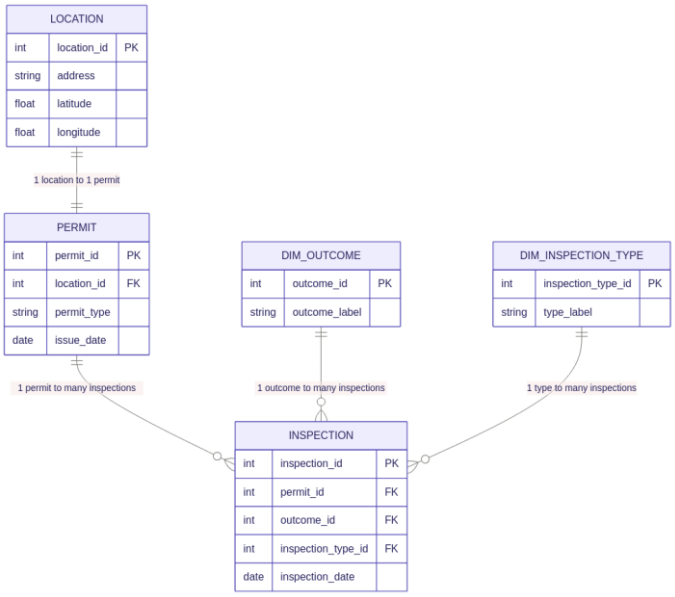
-- Exclude non-genuine outcomes
WHERE d.outcome_desc NOT IN (
    'Not Required',
    'Incorrectly Called For',
    'CSR Skipped',
    'Cancelled',
    'No Inspection Required',
    'Void')

-- Exclude records with no zone assignment
AND (
    l.address_line LIKE '% NW'
    OR l.address_line LIKE '% NE'
    OR l.address_line LIKE '% SW'
    OR l.address_line LIKE '% SE'
  )
ORDER BY i.inspection_completed_date;
```

4.2 Entity Relationship Diagram for our Dataset

To help understand how each of the tables used in our dataset is related, a five table crowfoot entity relationship model was designed to identify the structure and cardinality of our dataset.

Figure 4.1 Entity Relationship Diagram: *Calgary Building Inspections Table Join Structure*



***Adapted from the Calgary Building Inspections MySQL database. Model developed by (Group 1).*

4.3 Data Cleaning and Preparation Using Python

Following the export of the dataset from MySQL Workbench in CSV format, we imported our file into Python for preliminary inspection and data cleaning. Initial examination revealed that the dataset has a total of 105,462 rows across 10 columns. Data quality issues such as six records contained missing values in inspection_year and inspection_completed_date. These records were retained because those fields were not required for the Chi-Square test.

Secondly, "Failed" column contained 247 null values from invalid inspection types such as "*Not Required*, *Office Procedure*, and *CSR Skipped"*. Thirdly, the zone column was validated to confirm that all records corresponded to one of the four designated inspection areas.

In addition to cleaning, we converted the "failed" column to INT for the purpose of our regression model used in chapter 7. (See Python Data Exploration and Cleaning Code).

Chapter 5: Data Understanding

5.1 Data Summary

In this chapter, we explored and analyzed the cleaned dataset extracted in chapter 4 with 105,462 inspection records. Our aim is to understand the key metrics, distributions, and patterns in the data before any statistical testing is carried out. The table below summarize our data:

Metric	Value
Source database	Calgary Building Inspections (MySQL)
Tables joined	inspection, permit, location, dim_outcome, dim_inspection_type
Total inspections	Before: 107,067; After Cleaning: 105,462
Excluded outcomes	Not Required, Incorrectly Called For, CSR Skipped, Cancelled, No Inspection Required, Void
Control variable	inspection_type_desc (categorical)
Supplementary columns	latitude, longitude (for heatmap Visualization)
Average failure rate	36.86%
Number of zones	4 (NW, NE, SW, SE)
Number of inspection types	14 (Framing, Final, Foundation, etc.)
Date range	2000 to 2025 (concentrated in 2005 to 2023)
See Appendix: 5.1 Data Summary	

5.2 Geographical Distribution

This section presents the distribution of inspections by zone and evaluates the extent to which noncompliance rates differ across regions.

Zone	Total Inspections	Pass	Non-Pass	Failure Rate %
SE	33,411	21,177	12,234	36.62%
SW	28,880	18,380	10,500	36.36%
NW	25,754	15,937	9,817	38.12%
NE	17,417	11,093	6,324	36.31%
Total	**105,462**	**66,587**	**38,875**	**36.86%**
See Appendix 5.2: Geographical Distribution				

The results show that NW zone has the highest non-pass rate at 38.12%, which is 1.26 percentage points above the overall average of 36.86%. The NE zone has the lowest non-pass rate at 36.31%. The SE and SW zones are similar to each other.

5.3 Failure Rate by Inspection Type

In our research, inspection type is the control variable, and so we examined how failure rates vary across the 14 inspection types in the dataset. Our finding shows that inspection failure varies across types. Gas Final inspections had 82.4% failure rates as well as Gas Rough with 81.6% highest failure rates, while Foundation inspections had a 9.8% failure rate and Site Safety 9.5%. The difference between all four inspection types is significantly high compared to the 1.26 percentage points we have when comparing failure rates across zones. Whatever drives inspection failure rate in Calgary is more related to the type of job done than where it is done (See Appendix 5.3).

inspection_type_desc	inspections	Failure %
Gas Final	1841	82.4
Gas Rough	3895	81.59
FP Rough	1426	75.74
FP Final	19	47.37
Temp Gas	45	46.67
Final	21485	45.6
Framing	49675	38.07
Occupancy	3862	34.62
Pub Protect Safety	3	33.33
Underground	542	27.12
Rough	2383	26.23
Progress	9586	12.57
Foundation	10679	9.79
Site Safety	21	9.52

Table 5.3: *Total Inspections Failure Rate by Job Type*

5.4 Inspection Type Mix by Zone

If failure is highly clustered on inspection type, then the next question is whether the four zones receive the same mix of inspection work. The cross-tabulation below shows what share of each zone's inspections falls into the six most common types.

Type	NE (%)	NW (%)	SE (%)	SW (%)
Framing	48.07	52.22	44.72	44.70
Final	19.75	20.73	22.54	17.92
Foundation	9.93	10.00	10.59	9.82
Progress	9.53	4.90	9.54	12.04
Gas Rough	3.39	3.65	3.27	4.41
Gas Final	1.76	1.97	1.43	1.91

Table 5.4: *Inspection type mix by zone*

NW carries the highest share of Framing inspections at 52.22% versus roughly 44.7% in SE and SW; while Framing fails at 38.9%, which is above the city average. Also, NW carries the lowest share of Progress inspections at 4.9% against 9.5 - 12.0% elsewhere; while progress fails at only 12.0%. NW's is therefore prone to harder inspections and away from easier ones. Before concluding anything about geography, we need to ask how much of NW's elevated raw rate is simply this tilt.

5.5 Temporal Structure

Inspection numbers in Calgary have varied over the years. Volume surged to 10,038 inspections in 2007 and collapsed to 2,624 in 2008 following the global financial crisis, recovered to a second peak of 9,208 in 2015, and most recently fell from 7,220 in 2022 to 2,420 in 2023. Failure rates carry their own strong pattern, and it does not simply mirror volume.

Year	Inspections	Failure rate (%)	Year	Inspections	Failure rate (%)
2005	6,402	26.1	2015	9,208	36.7
2006	8,442	45.6	2016	8,536	30.9
2007	10,038	48.6	2017	4,412	40.5
2008	2,624	33.3	2018	4,682	43.2
2009	3,026	22.7	2019	2,789	42.2
2010	3,299	28.1	2020	2,036	27.6
2011	4,418	35.9	2021	3,195	25.3
2012	7,693	42.7	2022	7,220	30.0
2013	8,321	41.7	2023	2,420	31.1
2014	6,245	35.8			

Table 5.5: *Annual inspection volume and failure rate (years with at least 1,000 inspections)*

Across years with meaningful volume, failure rates range from 22.7% (2009) to 48.6% (2007) a variation of nearly 26 percentage points, far wider than anything the zones produce. The boom years of 2006–2007 averaged a 47.2% failure rate, consistent with construction quality suffering when activity outpaces skilled labor capacity, while the most recent period (2020–2023) averages 28.9%, suggesting improving compliance. This is a large, structured signal, and it motivates a modeling decision in Chapter 7: inspection year joins zone and inspection type as a predictor.

5.6 Findings from Data

The exploratory analysis revealed important patterns and trends within the dataset that will support the development and interpretation of our models. Key findings include:

- **The overall failure rate is 36.86%:** One in three inspections across Calgary results in a non-pass outcome.

- **The NW zone has the highest failure rate:** At 38.12%, the NW zone sits 1.26 percentage points above the overall average and 1.81 percentage points above the lowest zone, NE, at 36.32%.
- **Inspection Type differences are enormous.** Failure rates span roughly 73 percentage points across inspection types, confirming inspection type as the essential control variable.
- **Framing dominates the dataset:** Framing inspections account for 47.1% of all records, which is 49,678 out of 105,462 inspections recorded. The concentration of Framing inspections across zones may influence zone-level failure rates and is accounted for by including inspection type as a control variable in the regression model in Chapter 7.
- **Declining trend:** Failure rates peaked at 48.6% in 2007 before declining steadily to approximately 31.1% by 2023, suggesting improving compliance in recent years.
- **Time carries real signal.** Annual failure rates range from 23% to 49%, peaking in the 2006 – 2007 construction boom. This justifies adding inspection year to the predictive models.

Chapter 6: Data Visualization

Six visualizations were produced based on our findings from the previous chapter using Python (matplotlib, seaborn, and folium). Each addresses a distinct aspect of the research question. Together they all visually tell a story of how failure rates barely differ by zones, how inspection type differs enormously, how much each zone does different mix of work and how failure rates move with every construction cycle.

6.1 Failure Rate by Zone

Figure 1 shows the inspection failure across all four zone against the overall average of 36.86%. The NW zone has the highest failure rate at 38.12% which is 1.26 percentage points above the overall average. SE, SW, and NE cluster tightly between 36.32% and 36.61%.

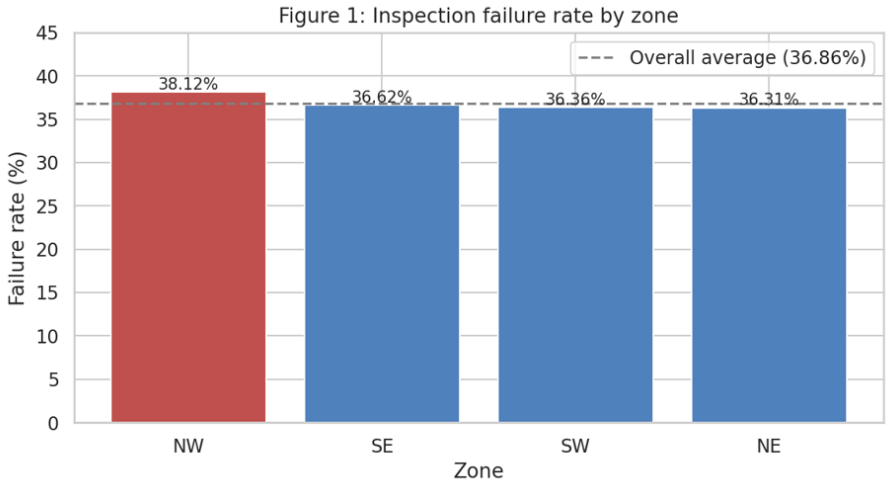


Figure 1: Inspection failure rate by zone. Dashed line = overall average (36.86%).

6.2 Pass vs Non-Pass by Zone

Figure 2 shows absolute counts. SE handles the most inspections (33,420) and NE the fewest (17,392), yet the failure proportions remain nearly identical across zones. NW's slightly elevated rate is proportional rather than a volume effect, and the similarity across all four stacks reinforces why formal statistical testing is needed to decide whether the small observed differences are meaningful.

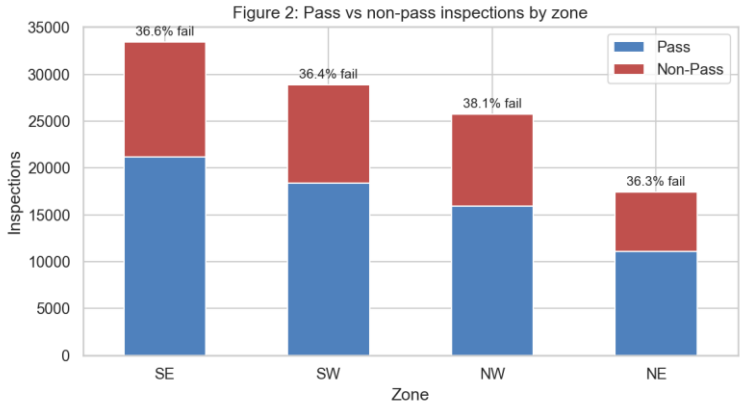


Figure 2: Stacked bar showing pass (blue) and non-pass (red) counts by zone. Percentages show each outcome's proportion within the zone.

6.3 Failure Rate by Inspection Type

Figure 3 makes failure rates across all 14 inspection types more visible. Two gas related inspections fail eight times out of ten when compared to foundation and site safety inspection which fails less than one time in ten inspections. For this reason, we would use the inspection type as a control variable in every model, as zones with more gas related inspections would have a higher failure rate.

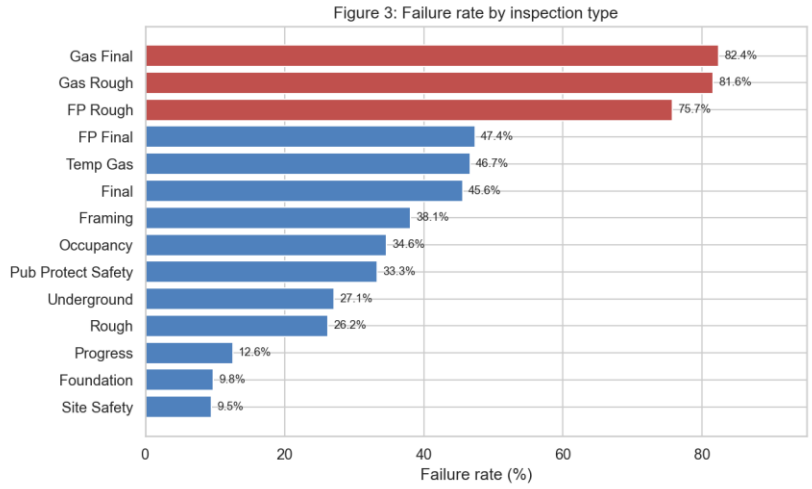


Figure 3: Failure rate by inspection type. Red Bars Exceeds 50% failure rate

6.4 Inspection Type Mix by Zone

Figure 4 visualizes the compositional structure from Table 5.4; the evidence at the heart of our argument. NW's Framing bar is visibly taller than every other zone's, and its Progress bar visibly shorter. This is the picture of a zone whose inspection portfolio is tilted toward harder work, and it is why NW's raw rate sits above the average in Figure 1.

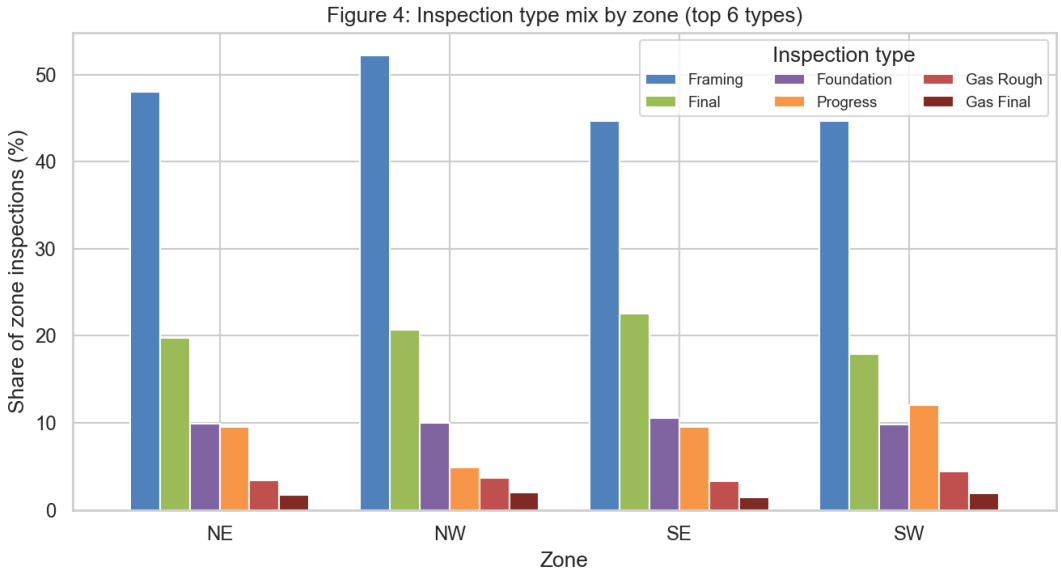


Figure 4: Share of each zone inspection by type (top six types)

6.5 Temporal Trends in Inspection Volume and Failure Rates

Figure 5 shows how inspection volume and failure rates changed across Calgary from 2003 to 2023 with construction and failure rate at its highest in 2007 followed by a sharp decline in 2008 during the financial crisis. Two metrics were plotted on the same chart using dual y-axes. The blue line tracks the total number of inspections per year on the left axis, while the red line tracks the failure rate percentage on the right axis.

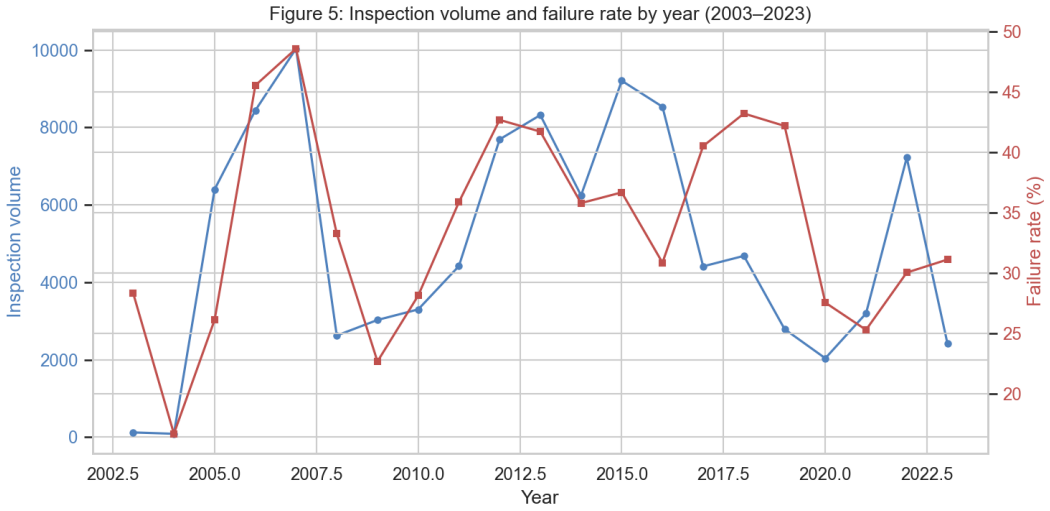


Figure 5: Inspection volume (blue, left axis) and failure rate percentage (red, right axis) from 2003 to 2023.

The chart reveals that Inspection volume peaked in 2007 at 10,038 before crashing to 2,624 in 2008 following the global financial crisis. A second peak of 9,208 occurred in 2015, with volume reaching 7,220 in 2022 before falling to 2,420 in 2023.

Failure rates do not mirror volume trends. The highest rate (48.61%) occurred during the 2007 boom, while the lowest (16.67%) was recorded in 2004. Between 2012 and 2019, failure rates remained persistently elevated (35–43%) despite fluctuating volumes, indicating that failure rates reflect construction quality and inspection standards rather than inspection frequency alone.

6.6 Geographic Distribution of Inspections Across Calgary Zones

Figure 6 maps a random sample of 10,000 inspections by GPS coordinates, colour coded by zone (blue = NW, red = NE, amber = SW, green = SE). Two black reference lines mark Calgary's north to south (51.047°N), and east to west (−114.057°) quadrant boundaries while each color cleanly aligns with each zone reference lines.

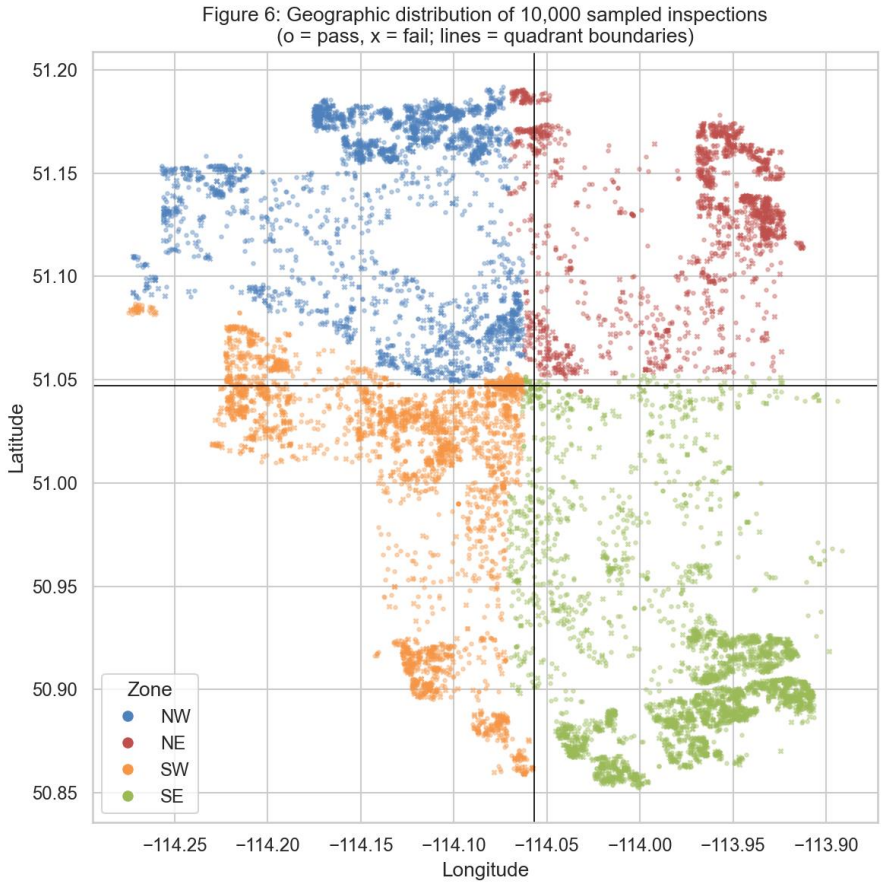


Figure 6: Geographic distribution of 10,000 random sampled inspections across Calgary zones. *Zone boundaries shown by black reference lines.*

Interpretation

The map shows that inspections are not evenly spread across the geographic area of Calgary. Southern zones (SE, SW) show a wider geographic spread reflecting newer suburban expansion, while northern zones cluster more tightly. Failed inspections are distributed throughout each zone rather than concentrated in specific sub-areas, confirming that failure is not location-specific within zones

The geographic boundaries between zones marked by the two black reference lines aligns closely with where the color of points changes, confirming that the address based zone assignment used throughout this analysis is geographically accurate and consistent with the actual GPS coordinates of each permit site.

6.7 Summary

Every visualization tells almost the same story which is zone failure rates are nearly indistinguishable (see Figures 1 and 2), inspection types vary enormously (Figure 3), zones differ in their mix of inspection work (Figure 4), and failure rates carry strong temporal structure tied to construction cycles (Figure 5), while geography within zones shows no failure hotspots (Figure 6). The modeling chapters now test formally what these figures suggest informally.

Chapter 7: Model Building

This chapter develops four analyses for increased flexibility. A Chi-Square test of independence, a logistic regression, a decision tree, and a random forest. The progression is deliberate. The Chi-Square test answers the hypothesis directly; logistic regression quantifies effect sizes; and the two tree-based models test whether the conclusion survives when we remove logistic regression's assumption that effects are linear and additive. If models with completely different structural assumptions all agree about the role of zone, the conclusion is triangulated rather than model dependent.

7.1 Data Preparation for Modeling

The modeling dataset is the 105,462 records analytical dataset established in Section 5.1. The target variable is failed, coded 1 for a non-pass outcome (36.86% of records) and 0 for a pass.

Three predictors were used. Zone and inspection type are the original research and control variables. Inspection year is added because failure rates were as low as 26.7% and high as 48.6% in other years. The difference between those two extremes is nearly 26 percent points meaning the year inspection took place could also affect its outcome and exempting it from our model makes it weaker. Zone and inspection type are one-hot encoded with NE and Foundation as reference categories (each being the lowest-failure level of its variable), producing 16 binary features; inspection year enters as a numeric feature, for 17 features in total.

A 70/30 train-test split with stratified sampling preserves the 36.86% failure proportion in both partitions, giving 73,823 training and 31,639 test records. Stratification is the only class-balance measure applied; the consequence of the 63:37 class split for failure detection at the default decision threshold is assessed in Chapter 8 and discussed as a limitation in Section 8.7.

Setting	Value
Final modeling records	105,462
Features after encoding	17 (15 one-hot + inspection year + intercept handling per model)
Reference categories	Zone: NE; Inspection type: Foundation
Train / test split	70% / 30%, stratified (73,823 / 31,639)
Random seed	42 (all models and splits)

Table 7.1: Modeling configuration shared by all models

7.2 Chi-Square Test of Independence

The Chi-Square test addresses the hypothesis in its purest form: are zone and inspection outcome statistically independent? A contingency table of observed pass and non-pass counts per zone is compared against the frequencies expected under independence, with significance assessed at $\alpha = 0.05$. Effect size is measured with Cramér's V, which matters here because with more than one hundred thousand records, even trivially small departures from independence will be statistically significant. We present finding in section 8.2.

7.3 Logistic Regression

Logistic regression is the standard interpretable model for a binary outcome with categorical predictors. Each coefficient converts to an odds ratio measuring how a predictor shifts the odds of failure relative to the reference category, holding the other predictors constant. which is exactly the "zone effect after controlling for inspection type" quantity our research question requires. Configuration: scikit-learn Logistic Regression, max_iter=2000, random_state=42. Selected coefficients:

Feature	Coefficient	Odds ratio	Interpretation
Gas Final	1.921	6.83	6.8× the odds of failure vs Foundation
Gas Rough	1.796	6.02	6.0× the odds of failure vs Foundation
Final	0.178	1.19	19% higher odds vs Foundation
zone_SE	0.020	1.02	Negligible: 2% higher odds vs NE
zone_NW	0.008	1.01	Negligible: 1% higher odds vs NE
zone_SW	−0.008	0.99	Negligible: 1% lower odds vs NE

Table 7.2: Key logistic regression coefficients and odds ratios (zone rows highlighted)

It is evident that the coefficient level of gas-related inspection types carry six to seven times the odds of failure relative to Foundation work, while all three zone coefficients sit within two percent of no effect.

7.4 Decision Tree

Logistic regression scores each predictor separately and adds the scores up. It cannot detect situations where two predictors interact — for example, if a particular inspection type failed at an unusually high rate in one specific zone. The decision tree has no such restriction and will automatically find those combinations if they exist in the data. "Unlike logistic regression which produces a table of numbers, the decision tree produces a visual flowchart of yes/no questions. You can read it directly to see which features the model considered important enough to split on, and in what order — making it self-explanatory evidence, not just a metric.

Configuration: DecisionTreeClassifier, max_depth=5 (keeping the tree interpretable), min_samples_leaf=200 (preventing overfit to tiny pockets), random_state=42.

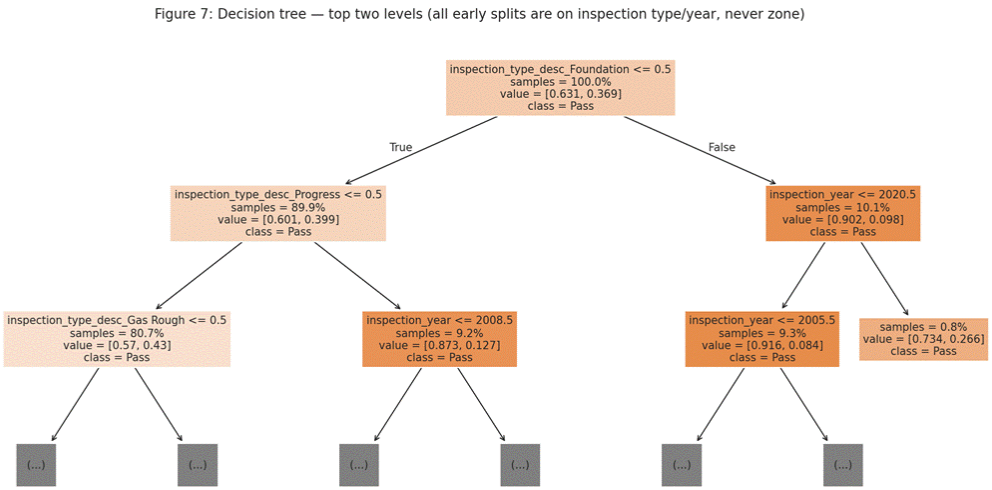


Figure 7: Top two levels of the fitted decision tree.

The fitted tree is itself evidence. Its first split is on Foundation inspections; the next levels split on Progress, Gas Rough, and inspection year. Zone features appear only in deep, low-importance branches. Given complete freedom to split on whatever predicts failure best, the tree essentially ignores geography.

7.5 Random Forest

A single decision tree may produce different results because small changes in the data can substantially alter its structure. Random forest addresses this by combining hundreds of trees, each trained on different bootstrap samples and random subsets of features. This reduces variance, improves stability, and provides an additional estimate of feature importance. For this reason, we used random forest to revalidate the decision tree results, as it is highly flexible and can capture complex, non-linear relationships. If random forest also ranks “zone” as having very low importance, this we conclude that zone contributes little to inspection failure rates, suggesting that the logistic regression findings are robust rather than simply a result of model limitations.

Feature	**Importance (%)**
Foundation	21.2
Inspection year	19.8
Gas Rough	17.8
Progress	16.4
Gas Final	9.0
FP Rough	4.6
Final	4.0
All zone features combined	1.4

Table 7.3: Random forest feature importance (top features; zone total highlighted)

Configuration: RandomForestClassifier, n_estimators=200, max_depth=12, min_samples_leaf=50, random_state=42.

The random forest results strongly suggest that inspection related features dominate predictive performance, accounting for approximately 79% of the model's total feature importance, while inspection year contributes another 19.8%. In contrast, the three zone variables combined account for only 1.4% of total importance. Because random forests can model complex non-linear relationships and interactions, these findings provide additional evidence that geographic zone contributes little predictive value in explaining inspection failure rates.

7.6 Model Selection Justification

All four analyses were not chosen as alternatives to each other, instead they were chosen because each one of them builds on the other and does a specific job.

- **The Chi-Square:** test our hypothesis, are zone and inspection outcome related? Cramér's V is added because with 105,462 records, even a meaningless association will appear statistically significant. Cramér's V cuts through that by measuring practical importance, not just statistical significance.
- **Logistic regression:** goes a step further by measuring exactly how much each zone raises or lowers the odds of failure, while holding inspection type constant.
- **Decision tree:** Removes the main limitation of logistic regression. its inability to detect interactions and provides a visual picture of which features the data actually considers worth splitting on.
- **Random forest:** builds on the decision tree by running 200 trees instead of one, making the importance scores reliable rather than a product of one unstable tree. It also serves as the final cross-check: if even the most flexible model finds zone unimportant, it reinforces its results

Together: the Chi-test decides the hypothesis, the regression measures it, and the tree-based models verify that the answer is not an artifact of any single model's assumptions.

7.7 Cross-Validation

As a stability check, each model was evaluated with stratified 5-fold cross-validation on the training partition (ROC-AUC): logistic regression 0.681 (±0.003), decision tree 0.669 (±0.004), random forest 0.720 (±0.004). The tight standard deviations indicate that the single held-out test evaluation in Chapter 8 is representative rather than a fortunate split.

Chapter 8: Model Evaluation

8.1 Evaluation Methodology

All models are evaluated on the same held-out test set of 31,639 records (30% of the data, stratified) that no model saw during training, using the standard 0.5 decision threshold throughout. Six metrics are reported. Accuracy is included for completeness but is the least informative metric here: with a 63% pass rate, a model that predicts "pass" for everything scores 63% accuracy while detecting zero failures. Because the classes are imbalanced, the threshold-independent metrics ROC-AUC (discrimination ability across all possible thresholds) and PR-AUC (discrimination weighted toward the minority failure class, the more demanding standard under imbalance); carry the most weight in the comparison, alongside precision, recall, and F1 at the default threshold.

8.2 Chi-Square Test Results

Statistic	Value
Chi-square statistic	23.78
Degrees of freedom	3
p-value	2.78×10^{-5}
Cramér's V	0.015
Decision ($\alpha = 0.05$)	Reject H_0

Table 8.1: Chi-Square test of independence (zone vs inspection outcome)

The Chi-Square test result ($p = 2.78 \times 10^{-5}$) tells us that zone and inspection outcome are related, so we reject the null hypothesis and support the alternate. However, this result needs context. With 105,462 records, the test is extremely sensitive and it will flag even the tiniest relationship as statistically significant, even if it does not actually matter in practice.

This is where Cramér's V = 0.015 becomes important. It measures the strength of the relationship rather than just its existence, and 0.015 is essentially zero on that scale. The biggest difference between any two zones is only 1.81 percentage points, and even that small gap is mostly explained by the fact that different zones handle different types of inspections.

Chi test reveals that zone and inspection outcome are statistically related, but that relationship is so weak it has no practical meaning. Knowing which zone an inspection took place in tells you almost nothing about whether it will pass or fail.

8.3 Comparative Model Performance

Table 8.2 presents all three predictive models on the same test set at the default 0.5 threshold.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic regression	0.674	0.807	0.152	0.256	0.677	0.557
Decision tree	0.667	0.853	0.116	0.204	0.670	0.635
Random forest	0.679	0.706	0.220	0.335	0.720	0.605

Table 8.2: Performance of all models on the held-out test set, default 0.5 threshold

Three patterns stand out. First, the random forest is the strongest model on every metric that balances both classes: the best F1 (0.335, against logistic regression's 0.256), the best recall (0.220), and the best ROC-AUC (0.720). Its ROC-AUC advantage over logistic regression and the single tree (0.72 versus 0.67–0.68) is genuine evidence of non-linear structure.

Second, every model is conservative at the default threshold: recall ranges from 0.116 to 0.220, meaning the majority of actual failures go unflagged. This is an expected consequence of the 63:37 class imbalance. A classifier minimizing overall error at a 0.5 cut naturally favours the majority pass class and it invariably affects all three models alike.

Third, the discrimination ceiling is real as no model exceeds 0.720 ROC-AUC, because zone, type, and year simply do not contain enough information to separate individual passes from failures more sharply. That ceiling is itself a finding: failure is driven substantially by job-level factors (contractor, workmanship, complexity) that this dataset does not record.

8.4 Confusion Matrices and Discrimination Curves

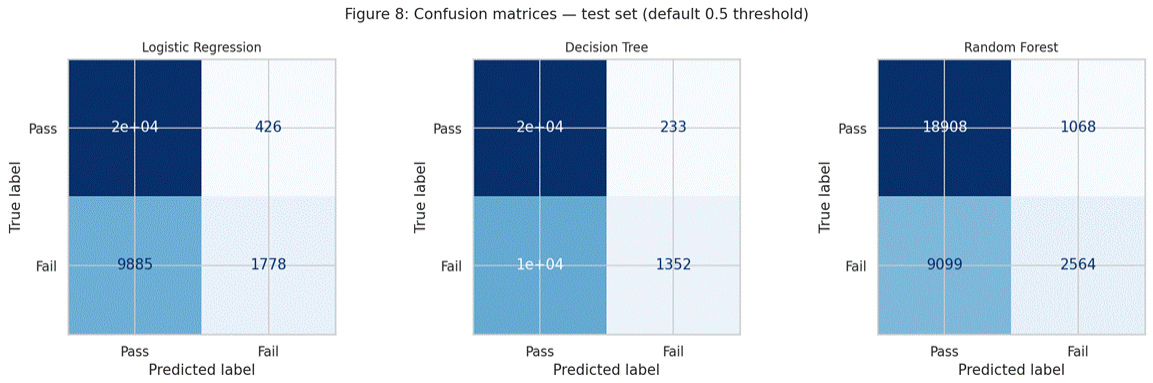


Figure 8: Confusion matrices for all three models on the test set (default 0.5 threshold).

The confusion matrices show the conservative pattern concretely. Of 11,663 actual failures in the test set, logistic regression flags 1,778 and the random forest 2,564 — the forest catches roughly 44% more failures than the regression at the same threshold, while both leave most failures

unflagged. The matrices also show why precision stays high: false positives are rare for every model (426 for logistic regression, 1,068 for the random forest, out of 19,976 actual passes).

Figure 9: Discrimination curves — test set

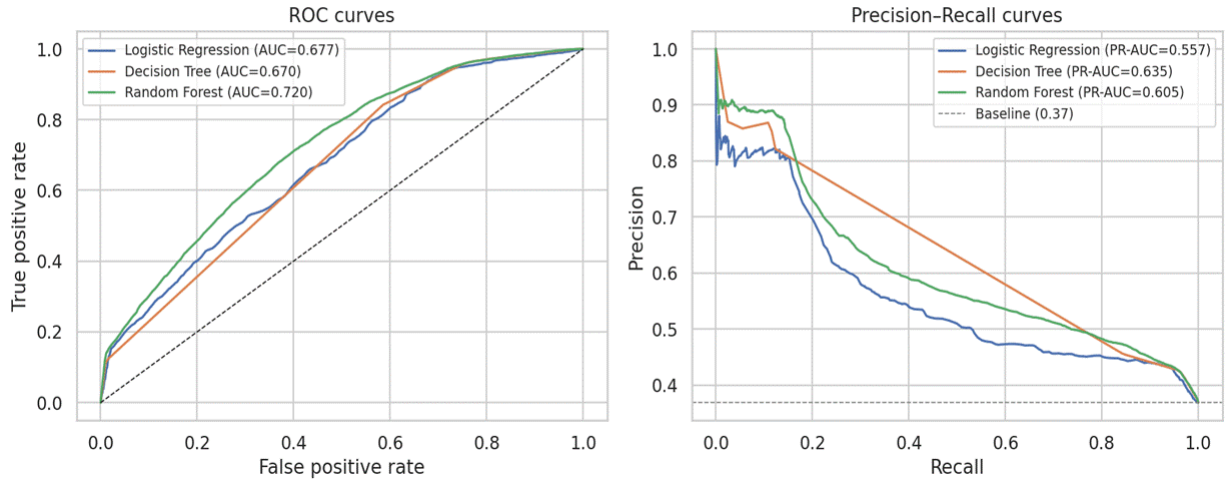


Figure 9: ROC curves (left) and precision–recall curves (right) for all three models.

The ROC curve graph shows random forest performing clearly better than logistic regression and the decision tree, confirming that flexible models pick up patterns the simpler models cannot detect. The precision-recall graph tells an equally important story — all models perform well above what random guessing would achieve, meaning they are genuinely learning to identify failure risk rather than just getting lucky. The key insight is that while all models become overly cautious when forced to make a hard yes/no decision at the 0.5 threshold, their underlying ability to rank inspections by failure risk remains useful and meaningful.

8.5 Feature Importance: the Triangulated Verdict on Zone

Figure 10: Random forest feature importance — zone features (red) vs type/year (blue)

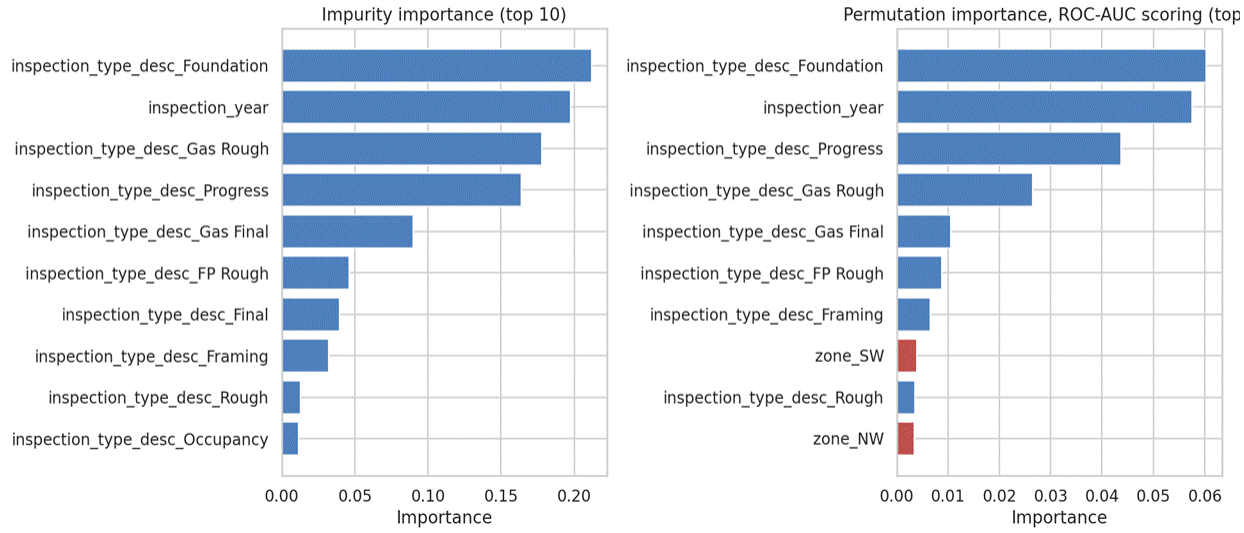


Figure 10: Random forest feature importance: impurity (left) and permutation with ROC-AUC scoring (right). Zone features in red.

This section brings together the strongest evidence in the study. Three completely different analytical methods — logistic regression, decision tree, and random forest — were each asked independently how much zone contributes to predicting inspection failure. They all reached the same conclusion: zone barely matters.

In logistic regression, zone coefficients are within 2% of having no effect at all. In the decision tree, zone was considered too unimportant to split on in the first three levels. In the random forest, zone accounts for only 1.4% of importance by one measure and 4.5% by another — and randomly scrambling zone information barely hurts the model's performance, while scrambling inspection type causes performance to collapse.

Each method has known weaknesses that a skeptic could raise. But those weaknesses don't all apply at the same time — each methods limitation is covered by another method's strength. When three fundamentally different approaches all agree, the conclusion is reliable: once inspection type and year are taken into account, which zone an inspection belongs to tells you almost nothing about whether it will pass or fail."

8.6 Interpretation of Findings

Hypothesis testing conclusion

The Chi-Square test rejects the null hypothesis of independence between zone and inspection outcome ($p = 2.78 \times 10^{-5} < 0.05$), formally supporting the alternate hypothesis. The complete answer, however, has two parts: the association is statistically real but practically negligible (Cramér's V = 0.015; maximum zone gap 1.81 percentage points), and the mix-adjustment analysis of Section 5.5 plus the triangulated feature-importance evidence of Section 8.5 show that what little zone-level difference exists is almost entirely a reflection of inspection type composition rather than geography.

What actually predicts failure

Inspection type: Gas-related inspections carry 6.0 to 6.8 times the odds of failure relative to Foundation work, consistent with the technical stringency of gas safety standards. Inspection year is the second pillar, contributing roughly a fifth of ensemble importance — failure rates climbed to 48.6% in the 2007 construction boom and have fallen to the high-20s since 2020, a pattern consistent with quality suffering when construction activity outpaces skilled-labour capacity.

Practical implication

For the City of Calgary, the evidence redirects attention: resources aimed at reducing inspection failures should target high-risk inspection types (above all, gas work) and boom-period workload pressures, not geographic zones. A zone-based deployment strategy would chase a 0.1 percentage-point residual while ignoring a 73-point type effect. The models' value for operations lies in their risk ranking rather than their default yes/no predictions: the random forest's failure probabilities (ROC-AUC 0.720) could prioritize re-inspection capacity toward the highest-risk inspections, with the operating threshold chosen to match available capacity.

8.7 Limitations and Implications

Model limitations

- **Low recall at the default threshold.** All models detect a minority of actual failures at the standard 0.5 cut (recall 0.12–0.22), an expected consequence of the 63:37 class imbalance. The comparison between models is unaffected since all are evaluated identically, and the threshold-independent metrics confirm the same ranking; but the default configurations are unsuitable as failure-screening tools without further work. Cost-sensitive class weighting or decision-threshold tuning would be the natural extension if a deployment required higher failure detection.
- **A real discrimination ceiling.** No model exceeds 0.720 ROC-AUC because the dataset omits the job-level factors — contractor identity and history, project complexity, permit

value, weather — that plausibly drive most failure variation. The models are honest about how much three administrative variables can explain.

- **Temporal signal is correlational.** Inspection year captures boom-bust effects but offers no causal account; year-based patterns may not extrapolate beyond the observed 2000–2023 window.

Data limitations

- **Four-zone granularity.** Quadrant-level zones may mask neighborhood level variation, though Figure 6 shows no visible failure hotspots within zones.
- **Excluded records.** 253 records (0.24%) were removed for administrative outcomes or missing dates; their exclusion is documented in Table 5.1 and is too small to affect any conclusion.

Implications for future research

First, adding more data such as contractor licensing records, permit values, or project types could improve prediction accuracy beyond the current 0.72 ceiling and test whether contractor quality, rather than geography, is the real driver of failure variation.

Secondly, the models should be validated temporally by training older data while testing on recent years to confirm that the patterns identified are stable over time and reliable enough for real-world use.

Finally, if these models were ever deployed operationally to screen inspections, the decision threshold should be adjusted and the model should be made more sensitive to failures because missing a genuinely deficient building carries far greater consequences than occasionally flagging a passing inspection for a second look.